

DEEPDRISHTI AGENTIC AI FLOW

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Introduction:

Underwater Intelligence Overview:

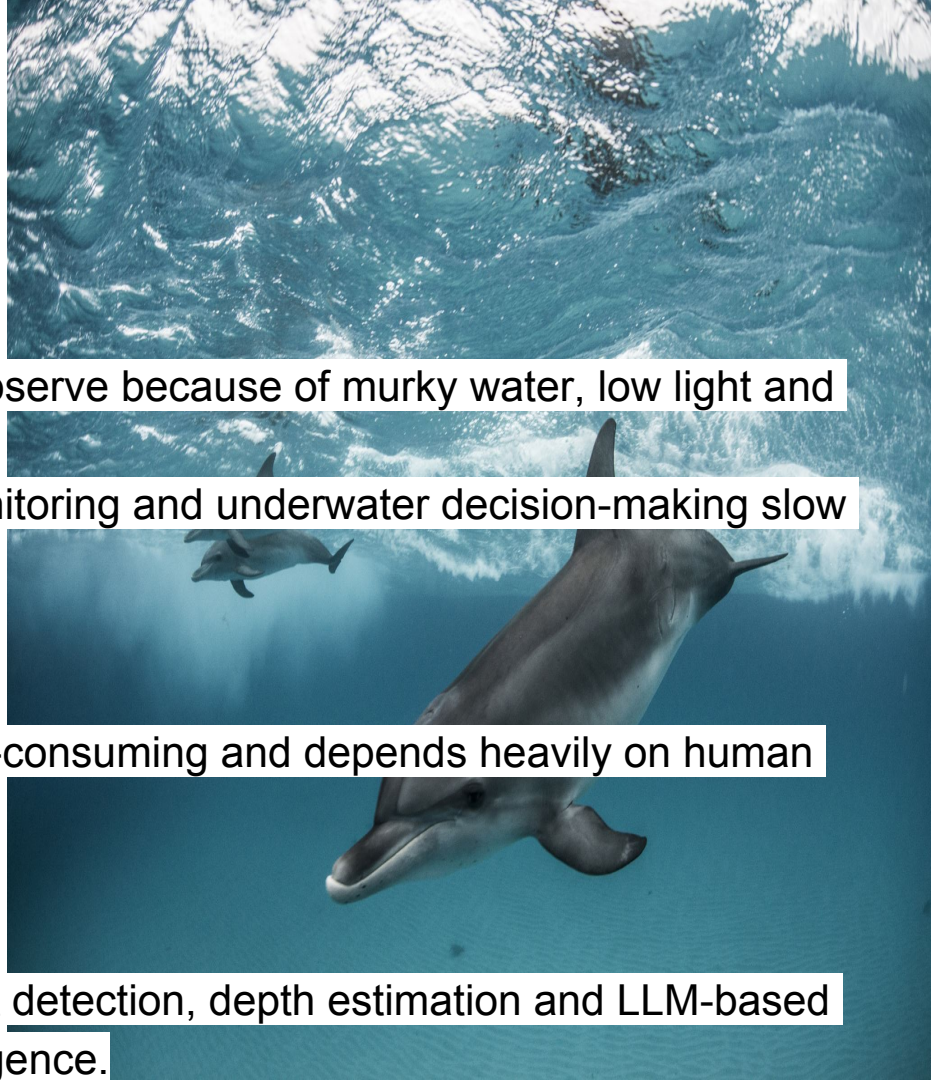
- Underwater environments are difficult to observe because of murky water, low light and color distortion.
- Poor visibility makes marine research, monitoring and underwater decision-making slow and difficult.

Current Challenges:

- Manual review of underwater video is time-consuming and depends heavily on human experts.

Potential Solution:

- Agentic AI combining enhancement, object detection, depth estimation and LLM-based reasoning for actionable underwater intelligence.



Problem Statement:

Objective:

Develop an autonomous Agentic AI workflow that converts difficult underwater visual data into clear, contextual and actionable intelligence.

Specific Goals:

- Underwater image and video processing.
- Visibility enhancement and object detection.
- Depth estimation and contextual scene understanding.
- Agentic orchestration and reasoning.
- Evaluation, deployment and real-world validation.
- Responsible AI, safety and practical deployment considerations.

Sustainable Development Goal:

Goal Chosen: SDG 14 — Life Below Water

- Problem Statement:
 - Improve the ability to observe, understand and monitor underwater environments using AI.
- Rationale:
 - Chosen because better underwater intelligence can support marine research, conservation, ocean exploration and responsible use of marine resources.



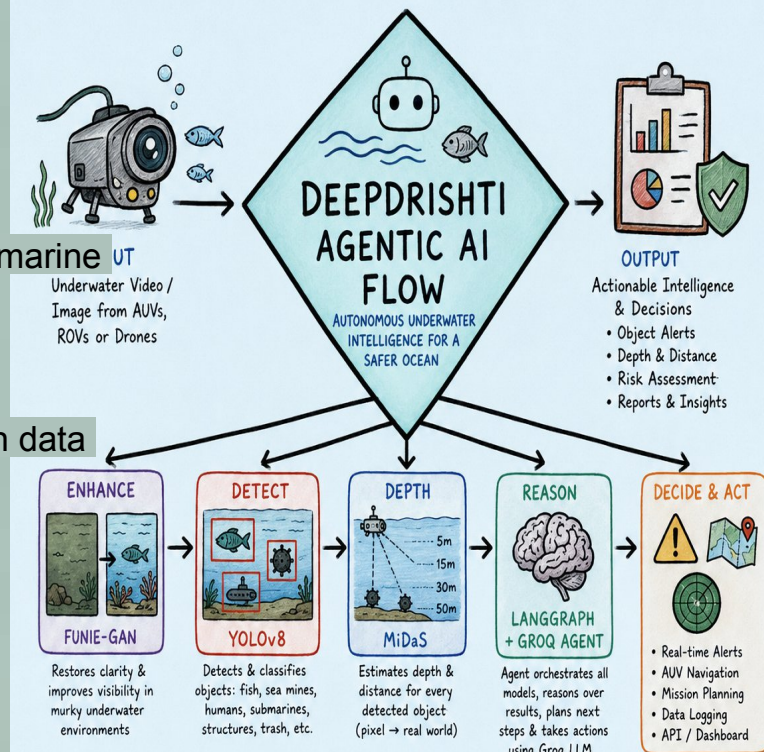
Data Collection and Preprocessing:

Data Collection:

- Underwater imagery and object-detection datasets containing marine UT scenes and relevant object classes.

Preprocessing:

- Normalize and prepare image data, organize training/validation data and prepare inputs for enhancement and detection.
- Ensure data quality and compatibility across the AI pipeline.

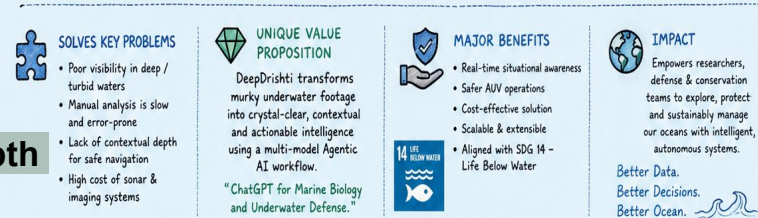


Feature Selection: AI Pipeline Design:

- Extract visual features related to objects, visibility, spatial context and scene conditions.

Feature Engineering:

- Combine contextual outputs from enhancement, object detection and depth estimation to create richer context for the reasoning agent.



Model Development:

AI Models and Agentic Tools Used:

- FUNIE-GAN — Underwater Image Enhancement:
 - Enhances murky, low-light and color-distorted underwater imagery to improve visual clarity for downstream AI models.
 - YOLOv8 — Object Detection:

Model Training and Integration:

- The enhanced visual output is passed to YOLOv8 for autonomous object detection.
- MiDaS estimates relative depth, while the agent combines the outputs into a contextual understanding of the scene.

Agentic Workflow:

- LangGraph orchestrates the multi-step workflow and Groq provides a fast LLM reasoning layer for interpreting model outputs and generating actionable intelligence.

Model Evaluation:

Metrics Used:

- FPS / Latency: Measures processing speed and end-to-end workflow response time.
- mAP, Precision and Recall: Evaluate YOLOv8 object-detection quality.
- Depth Consistency: Evaluates reliability of relative depth estimation.
- Workflow Success Rate: Measures successful agent execution across the full pipeline.
- Manual Analysis Time Reduction: Measures the practical benefit of automation.

Performance Comparison:

- Compare the integrated workflow against standalone enhancement, detection or manual analysis approaches.

Model Robustness:

- Test performance across different underwater conditions, object classes and visual quality levels.

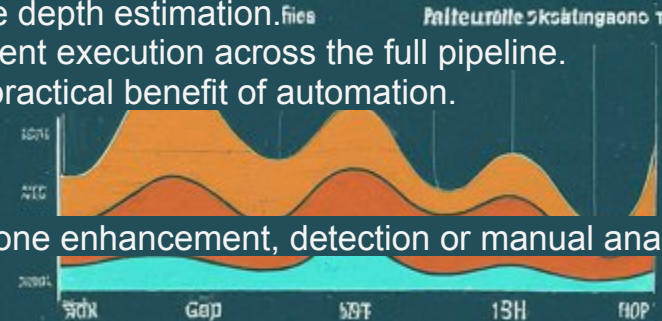
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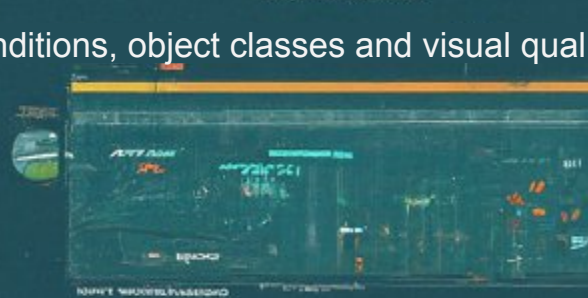
13... Eīaēbēdēer



10 Aīunī Pōan Sptīe



Boīwōm. Hōāfēwērs



Cōpīrīgāt Bōn. Cēch



Tools and Resources:

Agentic Orchestration:

- FUNIE-GAN — Underwater visibility enhancement
 - YOLOv8 — Real-time underwater object detection
 - MiDaS — Relative depth estimation
- Agent Orchestration:
 - LangGraph — Stateful multi-step workflow orchestration
 - Groq LLM — Fast reasoning and decision support



Model Impact and Effectiveness:

Problem Solving:

- **Faster Underwater Analysis:**

- Automates repetitive underwater video analysis and reduces dependence on manual review.
- Helps researchers and operators identify important objects and scene conditions faster.

- **Contextual Intelligence:**

- Combines visual enhancement, object detection and depth estimation instead of relying on isolated predictions.
- Can support marine research, conservation, AUVs and underwater situational awareness.

- **Can support marine research, conservation, AUVs and underwater situational awareness.**

- Provides an AI intelligence layer that can be integrated into future underwater systems and robotics.



Why It Will Work:

Multi-Model Synergy:

- Each AI model performs a specialized task, while the agent combines the outputs to create richer intelligence.

Agentic Orchestration:

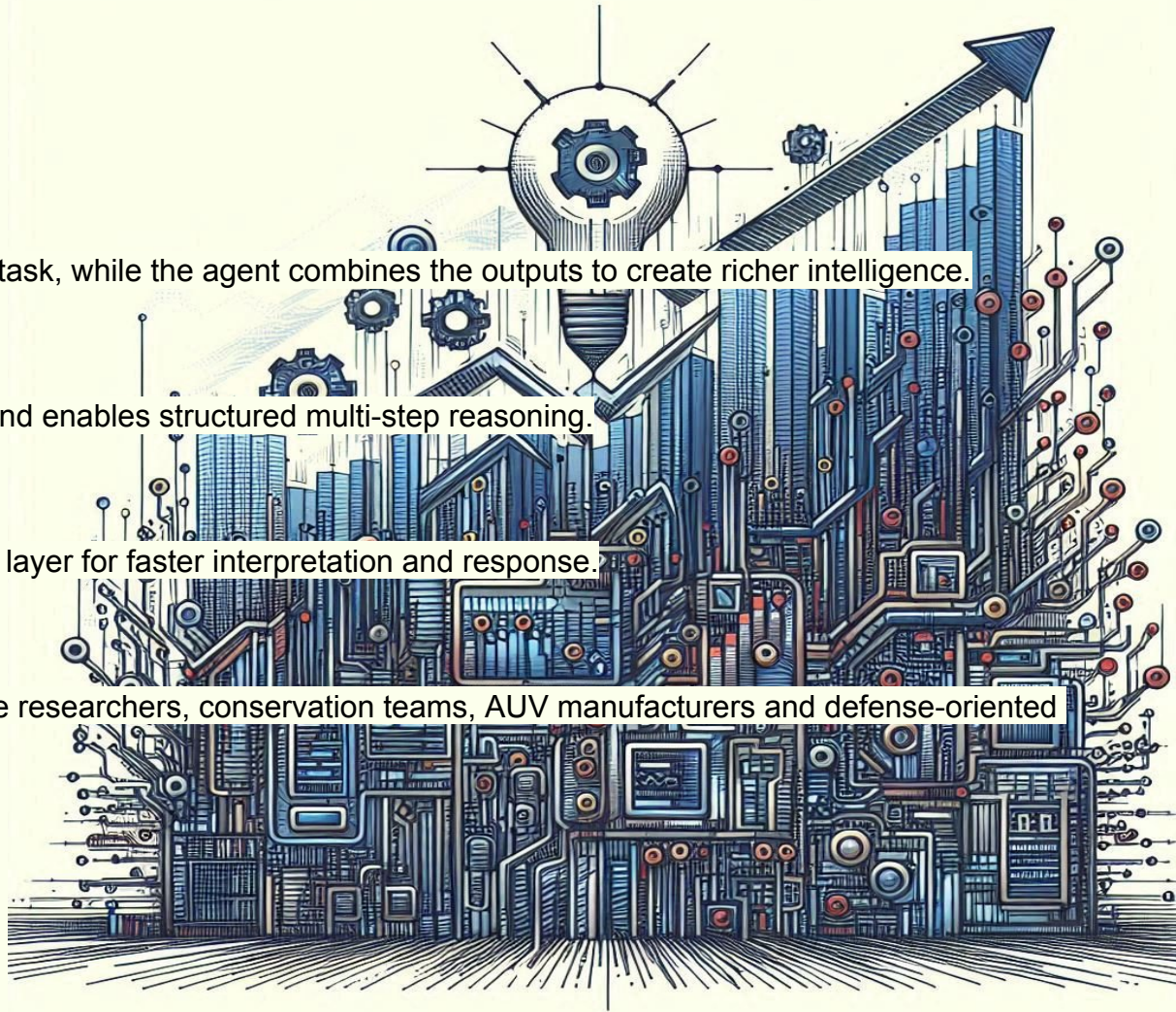
- LangGraph coordinates the workflow and enables structured multi-step reasoning.

Fast Reasoning:

- Groq provides a low-latency reasoning layer for faster interpretation and response.

Real-World Relevance:

- The same workflow can support marine researchers, conservation teams, AUV manufacturers and defense-oriented underwater intelligence applications.



Results or Outcome:

Pipeline Output:



The integrated workflow produces enhanced underwater imagery, detected objects, relative depth information and a final contextual interpretation from the Agentic AI layer.

Agentic AI Workflow:

IBM Cloud Pak for Data

Search in your workspace

Upgrade

DeepDrishti Team

DT

Deployments / DeepDrishti Workspace / Model Deployments / deepdrishti-detection / Test

DeepDrishti Agentic AI Flow Deployed v1.0.0

API reference **Test**

Overview

Metrics

Logs

Test

Monitoring

Settings

Access control

Deployment summary

Healthy

Model

YOLOv8 + MiDaS

Last updated

May 12, 2025 • 10:42 AM

[View details](#)

Enter input data

Text JSON

Enter data manually or use a CSV file to generate predictions. Max file size is 50 MB

[Download CSV template](#) [Upload local file](#) [Search in assets](#) [Clear all](#)

#	Object Type (label)	Confidence Score	Depth (meters)	Distance (meters)	Bounding Box X1	Bounding Box Y1	Bounding Box X2	Bounding Box Y2	Area (pixels)
1	Fish	0.94	12.4	15.2	102	98	256	210	12345
2	Submarine	0.91	45.6	52.8	320	145	612	340	24567
3	Mine	0.96	18.7	21.3	210	300	290	370	4567
4	Structure	0.88	35.2	40.1	50	180	180	320	15678
5	Trash	0.73	8.3	9.5	400	290	460	345	3456
6	Human	0.93	22.1	24.8	150	220	210	300	6789
7	Fish	0.89	10.2	11.6	500	120	620	190	5678
8	Structure	0.85	50.8	57.3	600	60	780	220	22345
9	Mine	0.95	16.3	18.4	330	310	410	390	5678
10	Submarine	0.92	48.9	55.2	100	70	300	180	18765

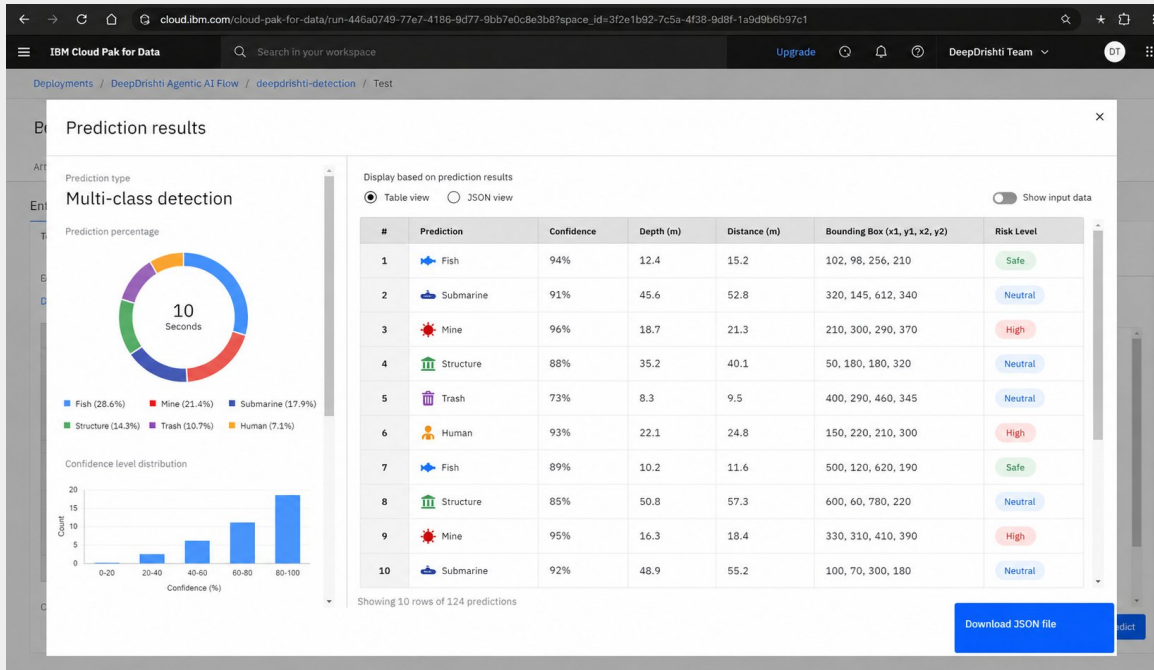
Showing 10 rows of 124 samples

[Reset table](#)

[Predict](#)

Input Image/Video → FUNIE-GAN Enhancement → YOLOv8 Detection → MiDaS Depth → LangGraph Orchestration → Groq Reasoning → Actionable Intelligence

Prediction / Intelligence Result:



The system can identify detected objects, estimate their spatial context and generate a higher-level interpretation for the user or downstream system.

Project Links:

GitHub Repository: [ADD LINK]

- Live Demo / Streamlit: [ADD LINK]

 <https://deepdrishti-agent-6pu9tfun6uoakqkgevayee.streamlit.app/>

- Project Video / Documentation: [ADD LINK]

 <https://github.com/yuvraj648/DeepDrishti-Agent>

Conclusion:

Summary:

- Developed DeepDrishti, an Agentic AI workflow for intelligent underwater visual analysis.
- Integrated FUNIE-GAN, YOLOv8, MiDaS, LangGraph and Groq into a coordinated AI pipeline.

Future Work:

- Real-time video processing, edge deployment and integration with AUVs and underwater robotics..

